Gokul Krishna J D

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.37
Velammal Vidyalaya Mel Ayanmbakkam Chennai TN	Class XII (Central Board of Secondary Education)	2023	93.60%
Velammal Vidyalaya Mel Ayanmbakkam Chennai TN	Class X (Central Board of Secondary Education)	2021	85.00%

INTERNSHIP

Summer Technical Intern | DevRev Inc., Bangalore

- **Data connector:** Built a metadata-sync connector for external warehouses (**Databricks, Snowflake, Microsoft Fabric**) that makes customer tables discoverable **without copying warehouse rows**, pulling data only when a query actually needs it -- reducing storage overhead and sync latency versus full replication.
- **AI agent:** Designed and built an **LLM-powered agent** that turns natural-language analytics requests into safe, reviewable data-ingestion workflows -- source discovery, query design, and refresh/schedule configuration, gated behind explicit approval for both job creation and first execution.
- **Evaluation harness:** Built a **20+ category benchmark suite** to quality-test the agent's behavior end-to-end: routing accuracy, schema/semantic correctness, cost-bounded query design, safety/PII exposure, and multi-turn revision, clarification, and destructive-action handling -- using a **weighted composite scorer** and a simulated-conversation pipeline for cases needing follow-up.
- **Reliability:** Integrated **Temporal** for durable async workflow orchestration with automatic retries, plus an active-cleanup failsafe for long-running ingestion jobs.
- **Catalog enrichment:** Built an **LLM-based enrichment pipeline** that auto-generates semantic descriptions for ingested schema metadata, indexed in **MongoDB** for the agent's discovery step.

Research Intern | IIT Hyderabad, under Dr. Sai Sidharth

- Formulated a novel **differentiable weight-matrix representation** for fractional-order derivatives, making non-local (memory-dependent) operators compatible with autograd-based training.
- Derived a **predictor-corrector numerical scheme** for non-linear distributed-order fractional differential equations, improving stability and convergence over standard methods for viscoelastic systems.
- Applied this formulation to build a **fractional-PINN framework** for viscoelastic mechanics -- implementation and results under *Projects*.

PROJECTS

Fractional-PINN | PyTorch, NumPy, Matplotlib

- Implemented the **weight-matrix formulation** in PyTorch, cutting numerical-differentiation compute time by **~1000×** and improving training convergence by **~100×** over standard finite-difference approaches.
- Built **PINN architectures** solving 1-DOF non-linear viscoelastic mechanics and structural beam vibration problems governed by fractional operators.
- Extended the framework into an **Inverse-PINN** pipeline to recover unknown material constants directly from noisy experimental vibration data -- validating the formulation against real measurements.

Hybrid Finite Element Analysis Tool | Python, C++

- Built a modular FEA solver (Abaqus-like) parsing .inp files to .vtu output -- Python for parsing/IO, C++ for the FEM solve -- with abstract mesh/element classes across 2D/3D and 4 element types; **faster than most peer submissions**.

Memory Management & Page Tables in GemOS | C

- Implemented custom **syscalls (mmap, munmap, mprotect)**, 4-level page-table fault handling, and a **Copy-on-Write** handler for shared read-only pages.

Multithreaded Sudoku Validator | C, C++, Pthreads

- Built a concurrent Sudoku validator using **POSIX threads, semaphores, and mutexes**, applying test-and-swap, compare-and-swap, and bounded compare-and-swap to prevent race conditions and deadlocks.

SKILLS

- **Languages:** Python, C, C++ (proficient); C#, TypeScript, Rust, SQL, Embedded C, LaTeX
- **Systems & Data Engineering:** MongoDB, REST APIs, POSIX Threads, Docker, workflow orchestration tools
- **ML & Scientific Computing:** PyTorch, NumPy, SciPy, Pandas, Matplotlib, BLAS
- **Tools:** Git/GitHub, Linux/Unix, Vim, Overleaf

RELEVANT COURSES

- **CS & AI Core:** Data Structures & Applications, Operating Systems I & II, Distributed Computing, Foundations of Machine Learning
- **Computational Core:** Fundamentals of Scientific Computing, Computer-Aided Numerical Methods I & II

POSITIONS OF RESPONSIBILITY

Core Member | Glitch Club, IIT Hyderabad

- Organised **technical sessions** on game development; grew club reach among students.

Team Member | Inter IIT Tech Meet 13.0, IIT Bombay

- Secured 6th among all IITs in a one-week game-jam; implemented NPC **behaviour and navigation** in Unity.

Core Member | Infero Club (Competitive Programming), IIT Hyderabad

- Mentored students in DSA; organised campus-wide contests.